Aim

To read student attendance and marks data from an Excel file, store the data in MongoDB, and retrieve students whose attendance is below 80%.

Description

This program reads student attendance and marks from an Excel file and uploads the data into MongoDB. It then finds and displays students whose attendance is below 80%.

Algorithm

- Read student data from the student.xlsx Excel file.
- Convert the Excel rows into student objects.
- Connect to MongoDB and store the student data in the attendance collection.
- Find students whose attendance is less than 80%.
- Display their details and the total number of students.

Program:

```
const { MongoClient } = require("mongodb");
const XLSX = require("xlsx");
const url = "mongodb://127.0.0.1:27017";
const client = new MongoClient(url);
const dbName = "studentDB";
const collectionName = "attendance";
const filePath = "D:\\student-grade-mongodb\\student.xlsx";
async function main() {
  try {
    const workbook = XLSX.readFile(filePath);
    const sheetName = workbook.SheetNames[0];
    const worksheet = workbook.Sheets[sheetName];
    const rows = XLSX.utils.sheet_to_json(worksheet, {
      header: 1
    });
    console.log("Excel data read successfully!");
```

```
console.log(`Total rows in Excel: ${rows.length}`);
const students = rows.slice(1).map(row => ({
    Rollno: Number(row[0]),
    Name: row[1],
    Attendance: Number(row[2]),
    Marks: Number(row[3])
}));
console.log("\nExcel students:");
console.table(students);
await client.connect();
console.log("\nConnected to MongoDB!");
const db = client.db(dbName);
const collection = db.collection(collectionName);
await collection.deleteMany({});
console.log("Old attendance data cleared.");
if (students.length > 0) {
    await collection.insertMany(students);
    console.log("Excel data uploaded to MongoDB!");
}
const lowAttendance = await collection.find({
    Attendance: {
        $lt: 80
    }
}).toArray();
console.log("\n=====================================");
console.log("STUDENTS WITH ATTENDANCE BELOW 80%");
console.log("=====================================");
if (lowAttendance.length === 0) {
    console.log("No students have attendance below 80%.");
```

```
    } else {
      lowAttendance.forEach(student => {
        console.log(`Roll No: ${student.Rollno}`);
        console.log(`Name: ${student.Name}`);
        console.log(`Attendance: ${student.Attendance}%`);
        console.log(`Marks: ${student.Marks}`);
        console.log("-------------------------------------");
      });
      console.log(
        `Total students below 80%: ${lowAttendance.length}`
      );
    }
  } catch (error) {
    console.error("\nError:");
    console.error(error);
  } finally {
    await client.close();
  }
}
main();
```

	A	B	C	D	E	F
1	Roll Number	Name	Attendance	Marks		
2	1	Krishna	93	90		
3	2	Koushik	90	91		
4	3	Suresh	75	92		
5	4	Vivek	79	93		
6	5	Anil	88	94		
7						
8						
9						
10						
11						
12						

Laboratory Record

of Full Stack Web Development (22CAC11N)

Roll No.: 160124748064

Experiment No.: 9

Sheet No.:

Date.: 08/10/2026

```
PS D:\student-grade-mongodb> node app1.js
Excel data read successfully!
Total rows in Excel: 6

Excel students:
```

(index)	Rollno	Name	Attendance	Marks
0	1	'Krishna'	93	90
1	2	'Koushik'	90	91
2	3	'Suresh'	75	92
3	4	'Vivek'	79	93
4	5	'Anil'	88	94

```
Connected to MongoDB!
Old attendance data cleared.
Excel data uploaded to MongoDB!

====================================
STUDENTS WITH ATTENDANCE BELOW 80%
====================================
Roll No: 3
Name: Suresh
Attendance: 75%
Marks: 92
------------------------------------
Roll No: 4
Name: Vivek
Attendance: 79%
Marks: 93
------------------------------------
Total students below 80%: 2
PS D:\student-grade-mongodb>
```

